

Sana Qureshi

Senior Backend Engineer | AI Systems

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PROFESSIONAL SUMMARY

Backend engineer with 6 years of experience building production AI systems and media infrastructure. Specialized in designing agentic workflows, LLM integration, and high-throughput processing pipelines serving millions of daily users. Deep expertise in cloud platforms, API architecture, and real-time data systems for entertainment and streaming contexts.

EXPERIENCE

Senior Systems Engineer | Lumina Media Labs | Portland, OR | March 2023 – Present

- Built and maintained core recommendation engine serving 12M+ daily active users across streaming platform, processing 500K+ content interactions per minute using multi-agent orchestration system
- Designed LLM-powered content tagging pipeline that processes 100K+ media assets monthly, reducing manual metadata work by 78% through autonomous agents evaluating title, genre, and content themes
- Implemented real-time API gateway handling 50K req/s peak traffic, managing rate limiting and request routing for downstream ML inference services with <50ms p99 latency
- Developed Python-based agentic framework for video transcription quality control, leveraging Claude API and custom reward models to validate outputs from 20+ transcription services in parallel
- Architected PostgreSQL and Redis infrastructure supporting stateful agent conversations, implementing distributed locking mechanisms to handle concurrent processing across 40-node Kubernetes cluster

Backend Engineer | Prism Digital Content | Seattle, WA | June 2021 – February 2023

- Engineered data ingestion pipeline for live broadcast metadata, processing 50M+ events daily from linear TV feeds into searchable catalog using Apache Kafka and Apache Spark
- Integrated AWS SageMaker for automated content classification, building ETL workflows that tagged 2M+ hours of archived content with semantic labels using LLM-based extraction
- Developed Python microservice for dynamic ad insertion logic, managing state across 10K+ concurrent user sessions with sub-100ms decision latency using Redis and gRPC protocols
- Implemented observability stack (Prometheus, Grafana, ELK) for media processing pipelines, tracking 200+ custom metrics across distributed system handling 2TB+ daily logs

Software Engineer | Nexus Entertainment Platform | Las Vegas, NV | August 2019 – May 2021

- Built backend services for interactive video platform supporting 3M+ concurrent viewers, designing stateless REST APIs and WebSocket handlers in Go for real-time event streaming
- Created automated content moderation system using OpenAI embeddings and cosine similarity, processing 500K+ user-generated clips daily with 94% accuracy against manual review baseline
- Optimized database queries and connection pooling, reducing p95 response times for metadata lookups from 800ms to 120ms and supporting 3x traffic increase without infrastructure scaling

TECHNICAL SKILLS

Languages & Frameworks: Python, Go, Java, SQL, TypeScript

Infrastructure & Cloud: AWS (Lambda, SageMaker, RDS, Kinesis), Kubernetes, Docker, Terraform

AI/ML Systems: LLM APIs, prompt engineering, embedding models, multi-agent orchestration, reward modeling

Data & Integration: Apache Kafka, PostgreSQL, Redis, GraphQL, gRPC, REST APIs, Elasticsearch